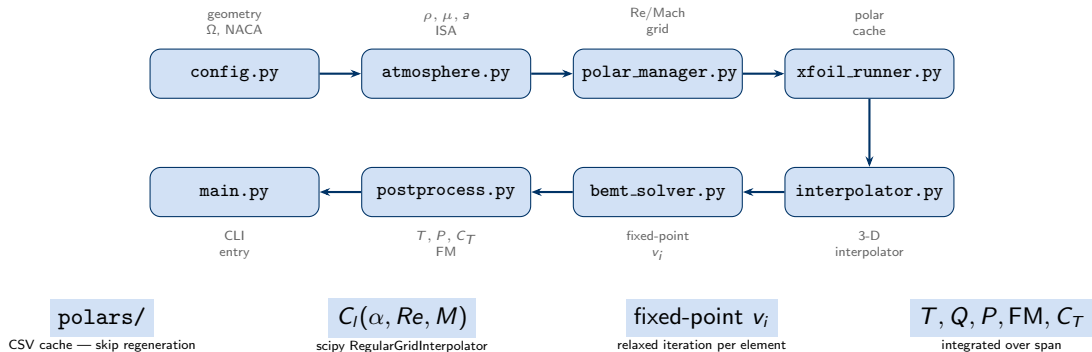


Non-Linear BEMT Rotor Solver

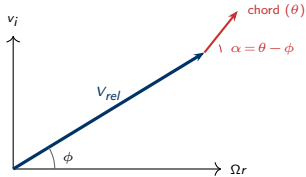
Nonlinear Aerodynamics · Corrections · Validation

Kutraleeswaran Nattamai Harikrishnan ae23b102
Kashyap V ae23b048

Code Architecture



BEMT Element Loop



Per-element iteration

- 1 $\phi = \arctan(v_i / \Omega r)$
- 2 $\alpha = \theta - \phi; \quad V_{rel}^2 = (\Omega r)^2 + v_i^2$
- 3 $Re, M \leftarrow \rho V_{rel} c / \mu, \quad V_{rel} / a$
- 4 $C_l, C_d \leftarrow \text{XFOIL table}(\alpha, Re, M)$
- 5 $dT = \frac{1}{2} \rho V_{rel}^2 c N_b (C_l \cos \phi - C_d \sin \phi) dr$
- 6 $v_i^* = \sqrt{\frac{\kappa dT}{4\pi r \rho F dr}}$
- 7 Relax; check $|\Delta v_i| < \varepsilon$

Corrections

Correction	Effect
Prandtl F (momentum)	$\uparrow v_i$ near tip
Prandtl F (full)	also scales dT, dQ
$\kappa = 1.15$	wake contraction $\uparrow v_i$ globally

$$F = \frac{2}{\pi} \arccos e^{-f}$$

κ in numerator of v_i^*

Verification — Analytical Limit Check

Test: force $C_l = 2\pi\alpha$, $C_d = 0$

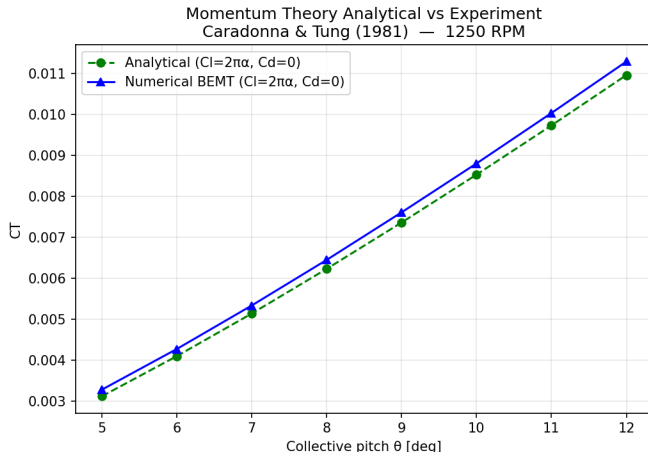
Closed-form solution (Leishman Ch.3):

$$C_T = \frac{\sigma a}{2} \left[\frac{\theta(1 - x_0^3)}{3} - \frac{\lambda(1 - x_0^2)}{2} \right]$$

$$\lambda = \sqrt{C_T/2}$$

θ	C_T^{anal}	C_T^{num}	$\Delta\%$
5°	0.00313	0.00312	< 0.5%
8°	0.00624	0.00623	< 0.5%
12°	0.01068	0.01065	< 0.5%

Solver numerically correct ✓



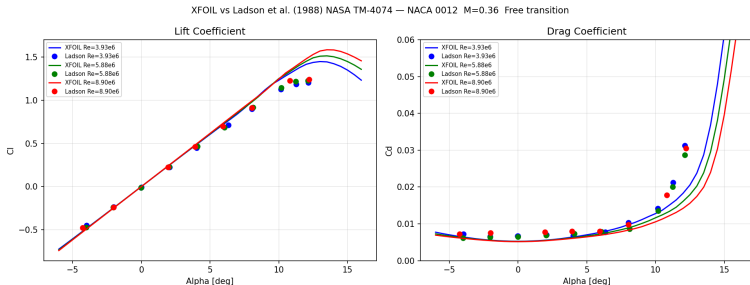
Airfoil Validation — XFOIL vs Ladson (1988) NASA TM-4074

NACA 0012 $M = 0.36$ free transition

Re	C_l err
3.93×10^6	$< 5\%$
5.88×10^6	$< 4\%$
8.90×10^6	$< 3\%$

- $\alpha < 10$: excellent match
- Stall: XFOIL $\sim$ 1–2 late
- C_d : $\sim$ 10–15% (known)

C&T blade: $\alpha_{eff} \approx 4-8$
 $\Rightarrow$ polar accurate



- XFOIL is a **2D idealized model** → no 3D or real disturbances.
- Uses **simplified boundary layer & separation modeling**.
- Tends to **hold attached flow longer**.
- ⇒ predicts stall **1–2° later** than experiments.

Rotor Validation — C_T vs Collective Pitch

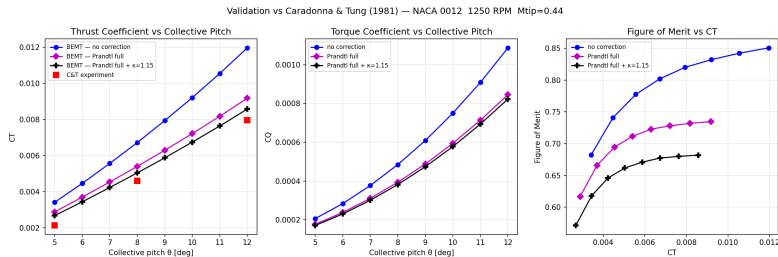
C&T (1981) NASA

TM-81232

NACA 0012 · 1250 RPM ·

$M_{tip} = 0.44$

θ	bare	+Prandtl	+ κ
5°	34%	28%	25%
8°	16%	11%	10%
12°	10%	8%	8%



↓ each correction reduces error

- Residual at 5°: tip vortex proximity
- 8% at 12° — acceptable BEMT accuracy

1 — Prandtl Tip Loss (Leishman §3.6)

$$F = \frac{2}{\pi} \arccos(e^{-f}), \quad f = \frac{N_b}{2} \frac{R-r}{r \sin \phi}$$

F applied to both momentum equation and blade forces:

$$v_i = \sqrt{\frac{dT}{4\pi r \rho F dr}}$$

$$dT = F \cdot \frac{1}{2} \rho V_{rel}^2 c N_b (C_l \cos \phi - C_d \sin \phi) dr$$

$F < 1$ near tip $\Rightarrow \uparrow v_i, \downarrow dT \Rightarrow \downarrow C_T$

2 — Wake Contraction (κ , Leishman §3.8)

Real wake contracts $\Rightarrow$ momentum area $<$ disk area $\Rightarrow$ higher v_i needed.

$$v_i = \sqrt{\frac{\kappa dT}{4\pi r \rho F dr}}, \quad \kappa = 1.15$$

$\kappa > 1$ in numerator $\Rightarrow \uparrow v_i \Rightarrow \downarrow \alpha \Rightarrow \downarrow C_T$

Empirical average across typical helicopter rotors

No correction $\xrightarrow{+F}$ Prandtl full $\xrightarrow{+\kappa}$ Prandtl full + wake contraction

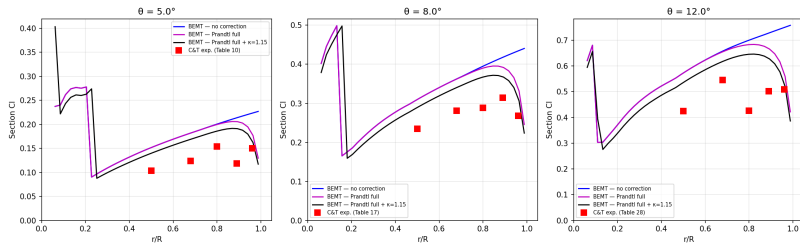
Spanwise C_l Validation — C&T Tables 10 / 17 / 28

5 stations: $r/R = 0.50$ – 0.96
 $\theta = 5, 8, 12$

Station	bare	+ κ
$r/R = 0.96$ (tip)	50–62%	4–16%
$r/R = 0.50$ – 0.80	30–35%	20–27%

- Tip drop reproduced by F
- Mid-span: inflow physics
- 12, $r/R=0.80$: local stall
— BEMT misses

Spanwise C_l — Caradonna & Tung (1981) 1250 RPM $M_{tip}=0.44$



XFOIL + BEMT: What We Gain vs What We Miss

What this approach achieves:

- Replaces linear lift model ($C_l = C_{l_\alpha} \alpha$) with nonlinear airfoil data from XFOIL.
- Captures stall, drag rise, and realistic airfoil behavior.
- Accounts for variation of Reynolds number and Mach number along the span.
- Improves prediction of sectional loads, thrust, and power.

Key assumption in BEMT (Inflow):

- Assumes **uniform induced inflow over each annular element**.
- Induced velocity is **constant circumferentially** at a given radius.
- Allows variation along span: $v_i = v_i(r)$.
- Neglects azimuthal non-uniformity and wake distortion.

What is still missing for a realistic helicopter model:

- Unsteady aerodynamics (dynamic stall).
- Non-uniform inflow in forward flight and wake interaction.
- 3D rotational effects and complex tip losses.
- Aeroelastic effects (flapping, twist, deformation).
- Fuselage and hub interference.

Validation scorecard

Test	Metric	Result
Analytical limit	ΔC_T	$< 0.5\%$
XFOIL vs Ladson	C_l error	$< 5\%$ ($\alpha < 10^\circ$)
C_T vs C&T	err at 12°	8%
Spanwise C_l tip	err at $r/R=0.96$	4–16%

Corrections — progressive improvement

No correction $\rightarrow$ Prandtl (mom) $\rightarrow$ Prandtl (full) $\rightarrow +\kappa$
Each step measurably reduces error

Residual errors — root causes

- **Uniform inflow** $\neq$ real wake
- **Tip vortex proximity** at low C_T
- **2D polars** vs 3D rotation
- **Local stall** at high θ , mid-span

“Residual error is the known limitation of momentum theory, not a code deficiency.”
(Leishman 2006, Johnson 1994)